Convexity-Based Approximation and Scaled Reduced-Form Model with Mass of Lower-Ranked Workers and Firm Wage Allocation

1 Core Idea: Convexity Approximation in the Non-Stochastic Model

In the original N-worker model, the lottery delivers a stochastic prize drawn from a Pareto distribution with heavy right tail. Low-position workers favor it because the utility function is convex below the reference point r ($\gamma_{\text{low}} > 1$), implying

$$\mathbb{E}[u(y)] > u(\mathbb{E}[y])$$

for random y with positive variance. Convexity amplifies rare large upside outcomes, creating effective risk-loving behavior despite a roughly mean-preserving lottery.

In the simplified non-stochastic model, we avoid repeated random draws. Instead, we replace stochastic lottery prizes with a deterministic equivalent that includes a linearized risk premium capturing the second-order welfare effect of variance via utility curvature.

The effective lottery benefit for a representative agent in class w includes a mean pool return share plus a risk-adjustment term

$$\rho_w = \frac{1}{2}u''(m_w) \cdot \widehat{\sigma}_w^2,$$

where a simple proxy for virtual variance is

$$\widehat{\sigma}_w^2 = c \cdot ((1-\lambda_w)m_w)^2 \qquad (c > 0 \text{ calibration constant}).$$

2 Scaled Reduced-Form Model with Mass at the Bottom

We consider a simplified two-class model with unequal population: one representative higher-ranked worker (mass 1) with income $m_H = R(1-\mu)$, and $W \gg 1$ identical lower-ranked workers, each with per-capita income

$$m_L = \frac{R\mu}{W}.$$

Each representative agent in class w chooses $\lambda_w \in [0, 1]$, the **fraction of per-capita income allocated to upskilling** (safe improvement channel). The fraction allocated to the shared lottery pool is therefore $1 - \lambda_w$.

The upskilling cost for a representative agent is linear in the upskilling share:

$$C_w(\lambda_w) = b\eta\lambda_w m_w,$$

where $b > 1$ approximates the original power-law friction, and $\eta \in (0, 1)$ is upskilling efficiency.

Aggregate lottery pool:

$$\Pi = W(1-\lambda_L)m_L + (1-\lambda_H)m_H.$$

Lottery return to a representative lower-ranked worker:

$$\kappa W(1-\lambda_L)m_L.$$

Lottery return to the higher-ranked worker:

$$\kappa(1-\lambda_H)m_H.$$

Net income per representative lower-ranked worker:

$$\begin{aligned} y_L = m_L + \eta\lambda_L m_L - C_L(\lambda_L) - (1-\lambda_L)m_L \\ + \kappa W(1-\lambda_L)m_L + \rho_L(1-\lambda_L), \end{aligned}$$

with $\rho_L(1-\lambda_L) = p_L(1-\lambda_L)^2$ and $p_L > 0$ (positive premium = bonus for lottery participation).

Net income for the higher-ranked worker:

$$\begin{aligned} y_H = m_H + \eta\lambda_H m_H - C_H(\lambda_H) - (1-\lambda_H)m_H \\ + \kappa(1-\lambda_H)m_H - p_H(1-\lambda_H)^2, \end{aligned}$$

with $p_H > 0$ (positive coefficient, but **subtracted** to create a penalty for lottery participation).

With linear utility $u(y) = y$, each maximizes own net income (Nash, treating other's choice fixed).

2.1 Low-class first-order condition (per representative low worker)

$$\frac{\partial y_L}{\partial \lambda_L} = (\eta - b\eta - \kappa W + 1)m_L - 2p_L(1-\lambda_L) = 0$$

Solving:

$$\lambda_L^* = 1 - \frac{m_L(\kappa W - \eta + b\eta - 1)}{2p_L},$$

bounded as $\lambda_L^* = \max\big(0, \min\big(1, \text{above}\big)\big)$.

2.2 High-class first-order condition

$$\frac{\partial y_H}{\partial \lambda_H} = (\eta - b\eta - \kappa + 1)m_H + 2p_H(1-\lambda_H) = 0$$

Solving:

$$\lambda_H^* = 1 - \frac{m_H(\kappa - \eta + b\eta - 1)}{2p_H},$$

bounded as $\lambda_H^* = \max\big(0, \min\big(1, \text{above}\big)\big)$.

3 The Firm Layer: Wage Allocation and Asset Optimization

The firm controls the total wage bill $R_t = \pi A_t$ (portion of current assets paid as wages). The firm chooses $\alpha \in [0, 1]$ such that: - Aggregate wages to lower-ranked workers (mass W) $= \alpha R_t$, - Wages to the higher-ranked worker (mass 1) $= (1 - \alpha)R_t$.

Per-capita wages are then $\alpha R_t/W$ for each lower-rank worker and $(1-\alpha)R_t$ for the higher-rank worker.

The firm maximizes the effective upskilling fraction of the wage bill:

$$\phi(\alpha) = \alpha\lambda_L^*(\alpha) + (1 - \alpha)\lambda_H^*(\alpha),$$

where $\lambda_L^*(\alpha)$ and $\lambda_H^*(\alpha)$ are the workers' equilibrium upskilling shares given policy α.

Asset dynamics:

$$\begin{aligned} A_{t+1} = (1 - \pi - \delta)A_t + \pi A_t \cdot \phi(\alpha) \\ - C(A_t) - K(\alpha_t - \alpha_{t-1}), \end{aligned}$$

with $C(A_t)$ (maintenance cost, e.g. cA_t^γ, $\gamma > 1$) and $K(\Delta\alpha)$ (adjustment cost of changing α, e.g. $k|\Delta\alpha|^\beta$, $\beta > 1$).

The firm chooses α_t to maximize A_{t+1} (or long-run asset value), anticipating workers' responses. This creates a two-level Nash equilibrium: - Workers optimize λ_L, λ_H given α. - Firm optimizes α given λ_L^*, λ_H^*.

The effective fraction $\phi(\alpha)$ is the share of wages that becomes productive investment. The firm typically sets high α^* (large wage share to lower ranks) when W is large, because the mass amplifies even modest upskilling from the bottom into large aggregate growth.